

Summary

Engineering Intern at Leap Finance building LLM pipelines, ML models, and data-driven products. B.Tech Mathematics & Data Science, NIT Bhopal (CGPA: 8.71). Skilled in Python, SQL, OpenAI APIs, LangChain, PyTorch, and BigQuery.

Experience

Leap Finance	<i>Jan 2026 – Present</i>
<i>Engineering Intern</i>	Bengaluru
- Contributed to Project Gangajal, a production LLM-based call scoring system (OpenAI, Python, LangChain) achieving 96.5% accuracy; supported integration with BigQuery as the data layer for score storage and reporting. - Scaled coverage to 100% of counseling calls by adding multilingual transcript translation (Hindi, Tamil, Telugu, Marathi) with LLM-based retry and error-recovery logic. - Shipped an early-warning system over student WhatsApp transcripts flagging drop-off signals; reduced counselor response time to at-risk students by 40%. - Worked across the Gangajal stack including Streamlit, Next.js, and FastAPI interfaces; wrote SQL queries for counseling funnel, visa tracking, and lead attribution reporting in BigQuery. - Built a LightGBM lead-scoring model on profile, session, and engagement data with SHAP explanations; helped counselors identify and prioritize high-intent applicants. - Scraped and structured a university database of 1,000+ institutions and 100K+ courses via Playwright, powering the College Finder product.	

Projects

Future Self	<i>Jun 2026</i>
<i>AI web app where three future versions of you respond to a life dilemma via parallel SSE streaming.</i>	[Live] [GitHub]
Tech Stack: Python, FastAPI, Groq API (LLaMA-3.3-70B), React, Vite, Supabase, Railway, Vercel - Built parallel SSE streaming with 3 simultaneous Groq LLM calls; future ages computed dynamically from user input to eliminate prompt contradictions. - Implemented multi-turn chat with persistent session memory per agent using Supabase; FastAPI on Railway, React/Vite on Vercel.	
AI Debate Arena	<i>Jun 2026</i>
<i>Live multi-agent debate platform where two AI personas argue any topic in real time over SSE streams.</i>	[Live] [GitHub]
Tech Stack: Python, FastAPI, Groq API (LLaMA-3.1-8B), Next.js, Supabase, Railway, Vercel - Architected multi-round debate flow with moderator summary; users vote on arguments round by round with results persisted in Supabase. - Deployed FastAPI backend on Railway and Next.js frontend on Vercel with full CORS and SSE streaming support.	
Sentiment Analysis using Deep Learning Models and Metaheuristic Optimizations	<i>Mar 2025</i>
<i>Built sentiment classifiers using BiLSTM-based deep learning with metaheuristic hyperparameter optimization.</i>	[Drive]
Tech Stack: Python, TensorFlow, Keras, NumPy, Pandas, Scikit-learn, NLTK, Word2Vec, FastText - Stacked BiLSTM + Conv1D layers with Attention and Residual Connections for richer sequence representations. - Cleaned and vectorized 2M+ samples via emoji normalization, slang expansion, and custom Word2Vec/FastText embeddings. - Tuned hyperparameters via Genetic Algorithm (GA) and Grey Wolf Optimizer (GWO): learning rate, dropout, and batch size. - Reached 95% accuracy on Amazon Fine Food Reviews and 85% on Sentiment140 (Twitter).	
Human Activity Recognition using CNN	<i>Apr 2024</i>
<i>Developed a CNN-based HAR system with transfer learning to classify 15 human activities from images.</i>	[GitHub]
Tech Stack: Python, TensorFlow, Keras, NumPy, Matplotlib - Benchmarked 5 CNN architectures — VGG, ResNet, GoogLeNet, DenseNet, MobileNetV2 — with transfer learning and augmentation on 12,000+ images, 15 activity classes. - Evaluated per-class performance using confusion matrices and accuracy curves across all architectures.	

Technical Skills

Languages & Web: Python, SQL, C++, R, MATLAB, HTML, CSS, Next.js, FastAPI
AI/ML & NLP: LLMs (OpenAI GPT-4.1), LangChain, Prompt Engineering, LightGBM, SHAP, PyTorch, TensorFlow, Keras, Scikit-learn
Data & Cloud: BigQuery, PostgreSQL, Playwright, Streamlit, Tableau
Tools: Git, GitHub, VS Code, Jupyter, Google Colab
Coursework: DSA, DBMS, Probability & Statistics, Linear Algebra, ML, Neural Networks

Education

Maulana Azad National Institute of Technology, Bhopal	<i>Aug 2022 – May 2026</i>
<i>B.Tech in Mathematics and Data Science</i>	CGPA: 8.71
Divya Bhaskar Public School	<i>May 2021 / May 2019</i>
<i>12th CBSE: 95.4% 10th CBSE: 91.4%, Bihar</i>	

Positions of Responsibility

Hospitality Lead	<i>May 2023 – Nov 2024</i>
E-Cell, MANIT – Led Central India’s largest hackathon (2,117+ participants, 25 finalist teams); oversaw logistics and speaker coordination for E-Summit’24.	
Rookie Member	<i>Feb 2023 – Jul 2023</i>
ISTE, MANIT – Organized 1-day TEDx talks; contributed to planning major inter-college tech events.	

Achievements

All India Rank 6 in IIM Sambalpur’s Ops’timato Case Study — Top 6 out of 769 teams.	<i>Mar 2025</i>
Qualified for National Round of Smart India Hackathon (SIH) 2024.	<i>Sep 2024</i>
Top 10 in Sahegi B-Plan Competition by IIM Ahmedabad (among 838 teams).	<i>Apr 2024</i>
PRMO Qualified in Class 10 (Mathematics Olympiad).	<i>Oct 2019</i>